

Analysis PhD Exam, May 2024

Write solutions in a neat and logical fashion, giving complete reasons for all steps.

Attempt SIX problems.

1. Consider Lebesgue measure on the real line $\mathbb{R}$. Give an example for each of the following, if possible. If not possible, give a brief explanation. A sequence f_n in $L^1(\mathbb{R})$ converging to an f in $L^1(\mathbb{R})$...
 - a) ...in the L^1 norm but not in measure,
 - b) ...in measure but not in the L^1 norm,
 - c) ...in the L^1 norm but not a.e.,
 - d) ...a.e. but not in measure.

2. State Tonelli's theorem, and give an example to show that the σ -finiteness hypothesis is necessary.
3. For σ -finite measures μ and ν , define what it means for μ to be absolutely continuous with respect to ν , and singular with respect to ν . State the Lebesgue–Radon–Nikodym theorem.
4. Let $(X, \mathcal{M}, \mu)$ be a σ -finite measure space. Suppose that g is a measurable function with the property that $gf \in L^2$ for all $f \in L^2$. Prove that $g \in L^\infty$.
5. Fix $1 < p < \infty$, let μ be a σ -finite measure, and let (f_n) be a sequence in $L^p(\mu)$. Suppose that there exists a function $f \in L^p(\mu)$ such that

$$\int f_n g d\mu \rightarrow \int f g d\mu$$

for every $g \in L^q(\mu)$, where $\frac{1}{p} + \frac{1}{q} = 1$.

- a) Prove that $\sup_n \{\|f_n\|_p\} < +\infty$.
 - b) Prove that $\|f\|_p \leq \limsup \|f_n\|_p$.
 - c) Give an example to show that strict inequality can hold in part (b).
6. Let $\mathcal{X}$ be a normed vector space. Say $x_n \rightarrow x$ weakly if $f(x_n) \rightarrow f(x)$ for all $f \in \mathcal{X}^*$. Prove that if $\mathcal{M}$ is a norm closed subspace of $\mathcal{X}$, and (x_n) is a sequence in $\mathcal{M}$ converging weakly to x , then $x \in \mathcal{M}$.
 7. Let μ be a finite signed Borel measure on $[0, 1]$. Suppose that

$$\int_0^1 \sin(2\pi nx) d\mu(x) = 0 \quad \text{and} \quad \int_0^1 \cos(2\pi nx) d\mu(x) = 0$$

for all $n = 0, 1, 2, \dots$. Prove that $\mu = 0$. Does it suffice to assume only the vanishing of the sine integrals? Prove, or give a counterexample.

8. Short answer.

- a) Give a brief explanation of how the Fourier transform is defined on $L^2(\mathbb{R})$.
- b) Sketch a proof that the Banach space $L^1(\mathbb{R})$ is not reflexive.
- c) Explain why the norm in $\ell^1(\mathbb{N})$ cannot be given by an inner product.